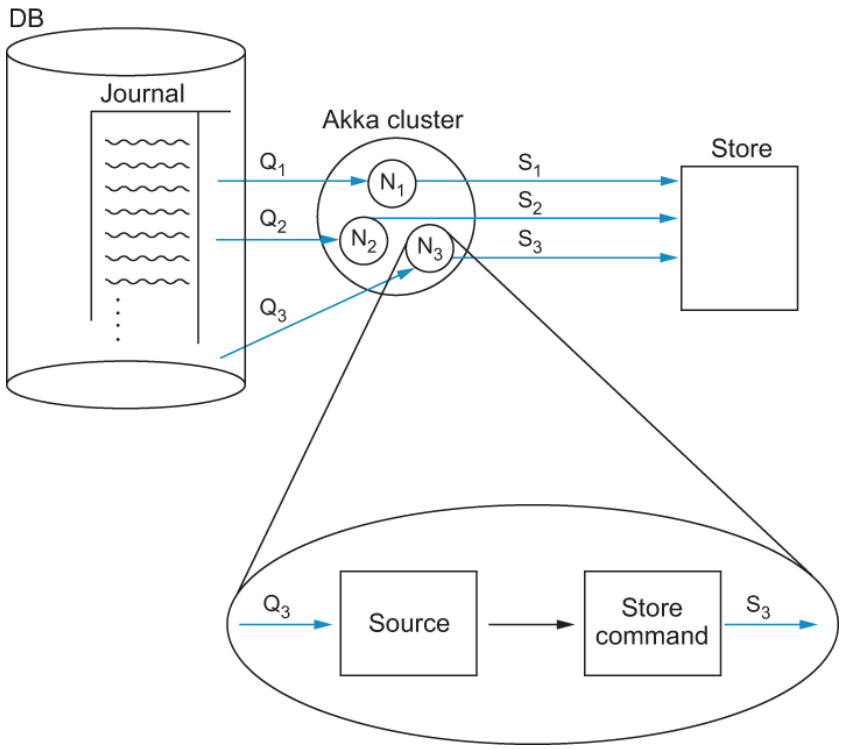


Figure 10.3 Generic case when reading from the journal

Clustering is not mandatory for persistence queries, but the cluster gives you fault tolerance and scalability, which you probably need anyway. The store in the upper-right corner can be a database or a topic. The `store` command is the connector: the producer responsible for writing to the store, upserting in the database, or committing to the topic.

DEFINITION An *upsert* behaves like an insert if there is no previous data to update or like an update if there is data already.

An addition to figure 10.3 would be to enrich the events with an external service after reading them. For example, you are only interested in cargo that contains certain materials. Depending on the contents, you may want to classify cargo as hazardous and treat it differently. With this information, the operator can make appropriate decisions at the docks—for example, to transfer hazardous cargo from one container to another or to prohibit it altogether.

An implementation using Akka Persistence Query first reads the journal, then filters the hazardous cargos, then calls an external service that adds some extra information about the related hazard, and finally writes them to a topic or database (see figure 10.4).